

CLF-C02 · BUILT FROM YOUR OWN TWO PDFs

The Only Document You Open Tomorrow

AWS Certified Cloud Practitioner — One-Sitting Mastery Guide

Merged and fact-checked against the official AWS Exam Guide (CLF-C02)

Everything in scope is here. Everything out of scope has been cut.

Built for a single 4–6 hour session, ending in a closed-book 50-question self-test.

How To Run Today: The One-Sitting Protocol

You said one document, one sitting, one pass, then book the exam. Here is how that actually works. This guide was built by cross-checking your **One-Day Prep Guide** (the detailed teaching content) against the **official AWS Exam Guide (CLF-C02)** (the authoritative scope document). Three things were done to it:

1. **Cut** — content that the official guide marks out-of-scope (e.g. CodeDeploy, CodeGuru, App Runner, Fraud Detector, Personalize, FSx for Lustre, Network Firewall) is flagged so you never waste a minute memorizing it.
2. **Filled** — services the official guide requires that your prep PDF barely mentioned (Amazon SES, AWS Amplify, AWS Audit Manager, AWS Global Accelerator, AWS PrivateLink, AWS License Manager, Service Catalog, Service Quotas, AWS Directory Service, AWS RAM, Elastic Disaster Recovery, AWS Client VPN, Migration Evaluator, Compute Optimizer, AWS CAF) have been added in short, exam-sized doses, marked **GAP-FILL**.
3. **Tested** — a 50-question closed-book self-exam at the end, weighted exactly like the real exam (24/30/34/12%), so you find out today — not tomorrow in the test center — what you actually know.

The Schedule (adjust start time, keep the ratios)

Block	What you do	Why this order
0:00 – 0:10	Read this page + skim the Table of Contents. Do not skip this — it's your map.	Orientation beats speed.
0:10 – 1:00	Domain 1 — Cloud Concepts (read once, slowly, out loud if you can)	Smallest domain, builds vocabulary you need for the rest.
1:00 – 2:15	Domain 2 — Security & Compliance (heaviest domain, 30%)	Biggest score chunk — give it the most focused hour.
2:15 – 2:25	Break. Stand up. Water.	Your brain needs it more here than anywhere else.
2:25 – 3:45	Domain 3 — Cloud Technology & Services (largest domain, 34%)	Most services, most tables — this is the volume domain.
3:45 – 4:15	Domain 4 — Billing, Pricing & Support + Comparison Tables + Numbers to Memorize	Smallest domain but easy, guaranteed points if drilled.
4:15 – 4:30	Exam Traps + Cheat Sheet + Exam Day Strategy (fast read)	Pattern recognition for the actual question wording.
4:30 – 5:15	Closed-book 50-question self-test. No peeking at earlier pages.	This is the real signal for "am I ready."
5:15 – 5:45	Grade yourself. Re-read <i>only</i> the sections behind your wrong answers.	Targeted repair, not a full re-read.
5:45 – 6:00	Skim Exam Day Strategy one more time. Book the exam if you scored 38/50+ ($\approx 76\%$).	700/1000 $\approx$ that ballpark with margin.

EXAM TIP: If, in the self-test, you score below 34/50 in any single domain's questions, do not book tomorrow's slot yet — reread that one domain block tonight and retake just those questions before you sleep. That is the only exception to "one document, one pass."

Table of Contents

1. Domain 1 — Cloud Concepts

24%

2.	Domain 2 — Security & Compliance	30%
3.	Domain 3 — Cloud Technology & Services	34%
4.	Domain 4 — Billing, Pricing & Support	12%
5.	Master Comparison Tables (the pairs the exam loves to confuse)	
6.	Numbers & Limits to Memorize	
7.	"Which Service Do I Use For...?" Cheat Sheet	
8.	20 Exam Traps & Wording Tricks	
9.	Explicitly Out-of-Scope — Stop Reading About These	
10.	Exam Day Logistics & Strategy	
11.	Final Gap-Fill Additions (from later source material)	
12.	Extra Concepts From Practice Exams & Tonight's Tests	
13.	Closed-Book 50-Question Self-Test + Answer Key	

Exam facts, straight from the official guide: Code CLF-C02 · 65 questions total (50 scored + 15 unscored, indistinguishable) · 90 minutes · Scaled score 100–1000, pass = 700 · Multiple choice (1 correct, 3 distractors) and multiple response (2+ correct out of 5+) · No penalty for guessing · Domain weights: Cloud Concepts 24%, Security & Compliance 30%, Cloud Technology & Services 34%, Billing/Pricing/Support 12%.

1.1 What Is Cloud Computing?

On-demand delivery of IT resources over the internet with pay-as-you-go pricing. No upfront hardware, no data-center maintenance.

The Six Advantages (know all six, and know them well enough to spot the odd one out)

1. **Trade CapEx for OpEx** — pay only for what you use, no upfront hardware investment.
2. **Massive economies of scale** — AWS's aggregate usage lowers per-unit cost for everyone.
3. **Stop guessing capacity** — scale up/down in minutes, eliminate over/under-provisioning.
4. **Increase speed & agility** — new resources in minutes, not weeks.
5. **Stop spending on data-center operations** — focus on what differentiates your business.
6. **Go global in minutes** — deploy to multiple Regions with a few clicks.

1.2 Service Models

Model	You manage	AWS manages	Examples
IaaS	OS, runtime, middleware, data, apps	Physical servers, network, storage	EC2, VPC, EBS
PaaS	Data & applications	Everything below the app layer	Elastic Beanstalk, RDS
SaaS	Nothing — just use the app	Everything	Gmail, Salesforce, Zoom

1.3 Deployment Models

- **Public Cloud** — all resources on AWS infrastructure, fully managed by AWS.
- **Private Cloud** — infrastructure operated solely for one organization, on-premises.
- **Hybrid Cloud** — mix of on-prem/private + public cloud via VPN or Direct Connect.
- **Multi-Cloud** — using more than one cloud provider simultaneously.

1.4 AWS Global Infrastructure

Region → Availability Zone (AZ) → Data Center → Edge Location

- **Region** — geographic area with 2+ AZs (e.g. us-east-1). Choose based on compliance, latency, service availability, and pricing.
- **Availability Zone (AZ)** — one or more discrete data centers with redundant power/networking, connected to sibling AZs by low-latency links. AZs do **not** share single points of failure. Deploy across 2+ AZs for High Availability.
- **Edge Location / Point of Presence** — used by CloudFront and Route 53 to cache content near users. More numerous than Regions.
- **Local Zone** — extension of a Region placed in a metro area for ultra-low latency.
- **Wavelength Zone** — AWS compute embedded inside telecom networks for 5G edge workloads.
- **AWS Outposts** — fully managed AWS infrastructure delivered on-premises.

EXAM TIP: Region = a city/country. AZ = one building in that city. Use multiple Regions for disaster recovery, business continuity, lower latency to distant users, or data sovereignty/legal requirements.

1.5 Cloud Adoption & AWS CAF GAP-FILL

The **AWS Cloud Adoption Framework (AWS CAF)** provides guidance to help organizations plan and execute cloud adoption. The official exam guide explicitly asks you to know its benefits, framed as: **reduced business risk, improved ESG (environmental, social, governance) performance, increased revenue, increased operational efficiency**. You just need to recognize this list — no deep detail required at this level.

Common migration strategies include database replication for live cutovers and using AWS Snowball for large offline data transfers ahead of a migration.

1.6 Cloud Economics: CapEx vs. OpEx

	CapEx (Traditional)	OpEx (Cloud)
Definition	Large upfront investment	Pay-as-you-go operational cost
Tax benefit	Depreciated over time	Deducted in the same year
Flexibility	Fixed capacity	Elastic, adjust anytime
Risk	High (over/under-provision)	Low (right-size as needed)

- **Rightsizing** — matching instance types/sizes to actual workload needs to avoid waste.
- **Licensing strategies** — Bring Your Own License (BYOL, use existing licenses) vs. AWS-included licenses (pay for license bundled into the service).
- **Automation benefit** — reduces manual effort and human error, freeing staff for higher-value work.
- **Economies of scale** — AWS's massive customer base drives down per-unit costs, passed on to you.

1.7 Well-Architected Framework — 6 Pillars

*Mnemonic: **OSRPCS** — Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, Sustainability.*

Pillar	Key idea	Example tools
Operational Excellence	Run & monitor systems to deliver business value; automate	CloudFormation, CodePipeline, Systems Manager
Security	Protect data and systems; least privilege everywhere	IAM, KMS, CloudTrail, GuardDuty
Reliability	Recover from disruptions; design for fault tolerance	Multi-AZ, Auto Scaling, Route 53 health checks
Performance Efficiency	Use computing resources efficiently	Right instance types, ElastiCache, CloudFront
Cost Optimization	Eliminate waste	Reserved Instances, Spot, Cost Explorer, right-sizing
Sustainability	Minimize environmental impact	Managed services, right-sizing, efficient Regions

AWS Well-Architected Tool (free, in-console) reviews your workloads against these six pillars and gives recommendations.

1.8 Cloud Architecture Design Principles

- **Design for failure** — assume things will fail, build redundancy.
- **Decouple components** — use SQS/SNS to separate tightly coupled services.
- **Implement elasticity** — Auto Scaling, serverless.
- **Think parallel** — horizontal scaling over vertical.

- **Stop guessing capacity** — provision dynamically.

2. Domain 2 — Security & Compliance

30%

This is the single biggest domain. Give it your sharpest hour.

2.1 Shared Responsibility Model

EXAM TIP: AWS is responsible **FOR** the cloud. YOU are responsible **IN** the cloud.

Layer	AWS Responsibility	Customer Responsibility
Physical	Data centers, hardware	Nothing
Network infrastructure	Global backbone	VPC config, security groups
Hypervisor	Managed by AWS	Nothing
OS (EC2)	None	Patch & harden guest OS
OS (RDS/Lambda)	AWS patches it	Nothing
Applications	None	Code, config, updates
Data	Encryption tools provided	Classify, encrypt, manage data
IAM	AWS IAM service itself	Users, roles, policies you create

Responsibility shifts by service: with EC2 you patch the guest OS yourself; with a managed service like RDS or Lambda, AWS patches the underlying OS for you. Recognizing this shift (EC2 vs. RDS vs. Lambda) is a favorite exam pattern.

2.2 Identity & Access Management (IAM)

IAM is **global** (not region-specific) and controls who can do what to which resources.

Core Concepts

- **Root Account** — email used to create the AWS account. Never use for daily tasks. Enable MFA immediately. Only the root user can do a small set of tasks (e.g. closing the account, changing account settings) — know that root protection is a distinct exam topic.
- **IAM User** — a person or application with long-term credentials.
- **IAM Group** — collection of users; apply policies to the group, not individuals.
- **IAM Role** — temporary credentials, assumed by users, services, or federated identities. Best practice for EC2 to access S3, etc.
- **IAM Policy** — JSON document defining Allow/Deny for Actions on Resources under Conditions.

Policy Evaluation Logic

1. Default: implicit **DENY** on everything.
2. Explicit **DENY** always overrides any Allow.
3. Explicit **ALLOW** grants access (unless there's also an explicit Deny).

IAM Policy Types

Type	Description
Identity-based	Attached to a user, group, or role
Resource-based	Attached to a resource (S3 bucket policy, SNS topic policy)

Service Control Policy (SCP)	Applied via AWS Organizations; limits max permissions for member accounts — does not itself grant permissions
Permission Boundary	Max permissions an identity can ever have; overrides identity-based policies
Session Policy	Passed when assuming a role via API

IAM Best Practices

- Enable MFA on root and privileged accounts.
- Grant least privilege — only what's needed for the task.
- Use roles instead of embedding access keys.
- Rotate credentials regularly; audit with the IAM Credential Report.
- Never share access keys; use IAM roles for EC2 instances.

Authentication & Federation GAP-FILL

- **AWS IAM Identity Center** (formerly AWS SSO) — single sign-on across multiple AWS accounts and business applications.
- **Federated identity** — letting users authenticate with an external identity provider (corporate directory, Google, Facebook) instead of creating an IAM user. **Amazon Cognito** is the service specifically for web/mobile app user identity (user pools + identity pools).
- **Cross-account IAM roles** — let a user/service in one account assume a role in another account without sharing credentials.

2.3 AWS Security Services

Service	Purpose & Key Facts
IAM	Authentication & authorization; global service
AWS Organizations	Manage multiple accounts; SCPs; consolidated billing
AWS Control Tower	Best-practice multi-account setup with guardrails
IAM Identity Center	Single sign-on for multiple accounts and apps
Amazon Cognito	User identity for web/mobile apps; user pools + identity pools
AWS KMS	Create and manage encryption keys; integrated with most AWS services
AWS CloudHSM	Hardware Security Module in your VPC; FIPS 140-2 Level 3; you manage the keys
AWS Secrets Manager	Store, rotate, retrieve database passwords and API keys automatically
AWS Certificate Manager (ACM)	Provision and manage SSL/TLS certificates — free for use with AWS services
AWS Shield Standard	Free DDoS protection, automatic on every AWS account
AWS Shield Advanced	Paid; 24/7 DDoS response team, financial protection, detailed attack metrics
AWS WAF	Web Application Firewall; filters HTTP(S) by rule; protects CloudFront, ALB, API Gateway
Amazon GuardDuty	ML-based threat detection; analyzes CloudTrail, VPC Flow Logs, DNS logs
Amazon Inspector	Automated vulnerability assessment for EC2 and container images
Amazon Macie	ML-powered discovery/protection of sensitive data (PII) in S3
AWS Security Hub	Centralized security alerts and compliance findings aggregator
Amazon Detective	Investigate security incidents; graph analysis of logs
AWS Firewall Manager	Centrally manage WAF rules, Shield Advanced, security groups across accounts
	Managed Microsoft Active Directory in AWS, or connect to an on-prem AD

AWS Directory Service

GAP-FILL

AWS RAM

GAP-FILL

Resource Access Manager — securely share AWS resources across accounts without duplicating them

AWS Audit Manager

GAP-FILL

Continuously audits AWS usage to assess risk and simplify compliance with regulations/standards

2.4 Logging & Auditing

Service	Answers the question	Notes
AWS CloudTrail	WHO did WHAT, when, from where?	Logs every API call. On by default for 90 days; create a Trail to S3 for long-term retention. CloudTrail Insights auto-detects unusual API activity.
Amazon CloudWatch	Is the system healthy right now?	Metrics, logs, alarms, dashboards. CloudWatch Logs Insights lets you query logs. EventBridge (formerly CloudWatch Events) reacts to events.
AWS Config	HOW did this resource's configuration change?	Records config changes over time; evaluates against compliance rules; shows compliance history.
VPC Flow Logs	What IP traffic hit my VPC/subnet/ENI?	Captures accept/reject traffic metadata.

2.5 Data Protection & Encryption

- **Encryption at rest** — KMS-managed keys (SSE-KMS), S3-managed keys (SSE-S3), customer-provided keys (SSE-C).
- **Encryption in transit** — TLS/SSL; ACM provisions the certificates.
- **S3 Object Lock** — WORM (Write-Once-Read-Many); Compliance or Governance mode.
- **S3 Versioning** — keeps multiple object versions; protects against accidental deletion.

2.6 Compliance Programs

AWS maintains compliance with PCI-DSS, HIPAA, SOC 1/2/3, ISO 27001, GDPR, FedRAMP, and many others.

- **AWS Artifact** — self-service portal to download compliance reports (SOC, PCI) and sign agreements (BAA, NDA). Free. This is a document portal, *not* a security scanning service.
- **AWS Compliance Center** — country-specific cloud compliance information.

3. Domain 3 — Cloud Technology & Services

34%

Largest domain. Highest volume of services. Do not try to memorize every feature — memorize what each service IS FOR, in one sentence.

3.1 Compute Services

Amazon EC2

Virtual machines. You choose OS, instance type, storage, networking.

Family	Type	Use case
t3/t4g	General purpose	Dev, test, low-traffic apps
m6i/m7g	General purpose	Web servers, databases
c6i/c7g	Compute optimized	HPC, ML inference, batch
r6i/r7g	Memory optimized	In-memory DBs, real-time analytics
p4/g5	Accelerated (GPU)	ML training, video rendering
i3/i4i	Storage optimized	NoSQL, data warehouses

EC2 Purchasing Options

Option	Discount	When to use
On-Demand	Baseline	Short-term, unpredictable, testing
Reserved Instance (RI)	Up to 72%	Predictable, 1 or 3 year commitment
Savings Plans	Up to 72%	Like RI but flexible (\$/hour commitment)
Spot Instance	Up to 90%	Fault-tolerant, flexible, batch jobs — can be interrupted
Dedicated Host	High	Compliance, BYOL
Dedicated Instance	Medium	Isolated hardware, avoid multi-tenancy
Capacity Reservation GAP-FILL	—	Reserve EC2 capacity in a specific AZ for any duration, no long-term commitment; combine with RI/Savings Plans for a discount

Key EC2 concepts: AMI = OS template. Security Group = stateful virtual firewall at instance level. Key Pair = SSH access. User Data = bootstrap script at launch. Instance Metadata = `169.254.169.254`.

Auto Scaling & Load Balancing

- **Auto Scaling Group (ASG)** — maintains a desired number of EC2 instances; scales out/in based on CloudWatch alarms (min/desired/max capacity). Recognize: **auto scaling = elasticity**.
- **Application Load Balancer (ALB)** — Layer 7 (HTTP/HTTPS); routes by path/host/headers; ideal for microservices.
- **Network Load Balancer (NLB)** — Layer 4 (TCP/UDP); ultra-low latency, millions of requests/sec, static/Elastic IP.
- **Gateway Load Balancer (GWLB)** — distributes traffic to virtual appliances (firewalls, IDS/IPS) at Layer 3.
- **Classic Load Balancer** — legacy, not recommended for new applications.

Serverless Compute

- **AWS Lambda** — run code without provisioning servers. Pay per invocation & duration. Max 15-min timeout. Integrates with 200+ services.
- **AWS Fargate** — serverless compute engine for containers (works with ECS and EKS). No EC2 management.

Container Services

- **Amazon ECS** — AWS-native container orchestration.
- **Amazon EKS** — managed Kubernetes.
- **Amazon ECR** — Docker image repository.

Other Compute

- **AWS Elastic Beanstalk** — PaaS; upload code, AWS handles deployment/capacity/LB/auto scaling; you still have access to underlying EC2.
- **Amazon Lightsail** — simple VPS for small apps/blogs/websites; fixed monthly price; bundles compute, storage, DNS, IP.
- **AWS Batch** — fully managed batch computing on EC2 or Fargate.
- **AWS Outposts** — AWS-managed hardware in your own data center.

SKIP — OUT OF SCOPE: AWS App Runner and AWS Wavelength are officially out of scope for CLF-C02 — recognize the names, don't study details.

3.2 Storage Services

Amazon S3

Object storage (flat namespace). Unlimited storage. Objects up to 5 TB. Buckets are Region-specific but bucket names are globally unique.

Class	Availability	Use Case
S3 Standard	99.99%	Frequently accessed; default choice
S3 Intelligent-Tiering	99.9%	Unknown/changing access; auto-moves between tiers
S3 Standard-IA	99.9%	Infrequent access; retrieval fee applies
S3 One Zone-IA	99.5%	Non-critical data; lower cost, single AZ
S3 Glacier Instant Retrieval	99.9%	Archive; millisecond retrieval
S3 Glacier Flexible Retrieval	99.99%	Archive; minutes to hours retrieval
S3 Glacier Deep Archive	99.99%	Long-term archive; 12–48 hr retrieval; cheapest

S3 features: Versioning (keep all versions), Lifecycle Policies (auto-transition between classes), Replication (CRR cross-region for DR, SRR same-region for compliance), Transfer Acceleration (upload via CloudFront edge locations), Multipart Upload (recommended >100 MB, required >5 GB), Pre-signed URLs (time-limited temp access), Static Website Hosting, Object Lock (WORM), Access Points.

Block & File Storage

- **Amazon EBS** — persistent block storage for EC2, tied to one AZ, like a hard drive. Snapshot to S3 for backup. Types: gp3 (general), io2 (high IOPS), st1 (throughput), sc1 (cold).
- **Amazon EFS** — managed NFS file system shared across multiple EC2 instances and AZs. Scales automatically. Linux only.
- **Amazon FSx for Windows** — managed Windows file server (SMB/NTFS/AD integration).
- **Instance Store** **GAP-FILL** — temporary block storage physically attached to the host; very high performance but data is lost on stop/termination.

SKIP — OUT OF SCOPE: Amazon FSx for Lustre is explicitly out of scope for CLF-C02.

Data Transfer & Hybrid Storage

- **AWS Storage Gateway** — hybrid cloud storage bridge: File, Volume, or Tape Gateway. On-prem apps connect to AWS storage. (Sometimes described as a "cached file system" on the official exam guide.)
- **AWS Snow Family** — physical devices for offline bulk data transfer: Snowcone (smallest, edge computing), Snowball Edge Storage/Compute (petabyte-scale, edge ML), Snowmobile (exabyte scale, literal truck).
- **AWS Backup** **GAP-FILL** — centralized, automated backup service across EC2, EBS, RDS, DynamoDB, EFS, and more; understand its use case is "one place to manage backup policies across services."
- **AWS Elastic Disaster Recovery** **GAP-FILL** — minimizes downtime/data loss by enabling fast, reliable recovery of on-prem or cloud apps into AWS.

3.3 Database Services

Service	Type	Best For
RDS	Relational (SQL)	MySQL, PostgreSQL, MariaDB, Oracle, SQL Server; managed; Multi-AZ
Aurora	Relational (SQL)	MySQL/PostgreSQL-compatible; ~5x faster; 6 copies across 3 AZs; has a Serverless option
DynamoDB	NoSQL (key-value)	Single-digit millisecond latency; serverless; unlimited scale
Redshift	Data warehouse	Columnar; analytics/BI; petabyte scale; Redshift Spectrum queries S3 directly
ElastiCache	In-memory cache	Redis or Memcached; sub-millisecond latency; relieves DB load
Amazon Neptune	Graph database	Social networks, knowledge graphs, fraud detection
Amazon DocumentDB	Document (NoSQL)	MongoDB-compatible; JSON data

- **RDS Multi-AZ** — synchronous standby in a second AZ, automatic failover; for High Availability, **not** read scaling.
- **RDS Read Replicas** — asynchronous copy, for read scaling; can be cross-region.
- **Decision rule:** use an EC2-hosted (self-managed) database only if you need full OS-level control; otherwise use a managed AWS database service.
- **AWS DMS** — Database Migration Service; migrate on-prem or cloud DBs to AWS, homogeneous (MySQL → RDS MySQL) or heterogeneous (Oracle → Aurora).
- **AWS SCT** — Schema Conversion Tool; converts schema between different DB engines.

SKIP — OUT OF SCOPE: Amazon QLDB (ledger DB), Amazon Keyspaces (Cassandra), and Amazon MemoryDB for Redis are not on the official in-scope list — don't spend time on them.

3.4 Networking Services

Amazon VPC

Your own isolated network in AWS. You control IP ranges (CIDR), subnets, route tables, internet gateways, NAT, security groups, and NACLs.

- **Subnet** — range of IPs within a VPC. Public subnet = route to Internet Gateway. Private subnet = no direct internet route.
- **Internet Gateway (IGW)** — two-way internet access for public subnet resources.
- **NAT Gateway** — lets private subnet resources initiate outbound internet traffic (not inbound); managed by AWS; sits in a **public** subnet.

- **Security Group** — stateful firewall at instance/ENI level. Allow rules only; return traffic auto-allowed.
- **Network ACL (NACL)** — stateless firewall at subnet level. Allow and Deny rules; both inbound and outbound must be defined.
- **VPC Peering** — connects two VPCs privately; non-transitive (A ↔ B and B ↔ C does not mean A ↔ C).
- **AWS Transit Gateway** — hub-and-spoke for thousands of VPCs and on-prem networks; transitive routing.
- **VPC Endpoints** — private connection from VPC to AWS services without internet. Interface endpoint (most services, uses ENI, costs money) vs. Gateway endpoint (S3 and DynamoDB only, free).
- **Elastic IP** — static public IPv4 address, held until you release it.
- **AWS PrivateLink** **GAP-FILL** — privately connects your VPC to supported AWS services, your own services, or partner services, as if they were in your VPC — traffic never traverses the public internet.

DNS & Content Delivery

- **Amazon Route 53** — managed DNS. Routing policies: Simple, Weighted, Latency-based, Failover, Geolocation, Geoproximity, Multi-value. Also does domain registration.
- **Amazon CloudFront** — global CDN, caches content at 400+ edge locations, reduces latency, absorbs DDoS, integrates with Shield & WAF.
- **AWS Global Accelerator** **GAP-FILL** — uses the AWS global network (not the public internet) to route user traffic to the optimal healthy endpoint, improving performance for non-HTTP and multi-Region apps. (Difference vs. CloudFront: CloudFront caches content; Global Accelerator improves network path performance, works for TCP/UDP too.)

Hybrid Connectivity

- **AWS Site-to-Site VPN** — encrypted tunnel over the internet between on-prem and VPC. Quick to set up; variable latency.
- **AWS Direct Connect** — dedicated private physical connection (1/10/100 Gbps) from on-prem to AWS. Consistent latency, bypasses internet, takes weeks to provision.
- **Direct Connect + VPN combined** — Direct Connect for bandwidth, VPN for encryption — a common "highly secure hybrid connectivity" exam answer.
- **AWS Client VPN** **GAP-FILL** — managed client-based VPN so individual users/devices can securely connect to AWS or on-prem resources from anywhere.

3.5 Management & Monitoring

- **Amazon CloudWatch** — metrics, alarms, dashboards, log groups; Logs Insights to query logs; EventBridge to react to events.
- **AWS CloudTrail** — logs all AWS API calls; store to S3; detect suspicious activity.
- **AWS Config** — tracks resource configurations over time; evaluates against rules; shows compliance history.
- **AWS Trusted Advisor** — checks across 5 categories: Cost Optimization, Performance, Security, Fault Tolerance, Service Limits. Free tier gets 7 core checks; Business/Enterprise support gets all checks.
- **AWS Health Dashboard** — Service Health (public) + Account Health (personal) notifications; the AWS Health API lets you programmatically monitor account/service health for cost and operational management.
- **AWS CloudFormation** — Infrastructure as Code using JSON/YAML templates; deploys a "stack" of resources; repeatable, version-controlled; no extra charge beyond the resources created.
- **AWS Systems Manager** — manage EC2 at scale: Session Manager (no SSH needed), Parameter Store (config/secrets), Patch Manager, Run Command, Automation.
- **AWS Auto Scaling** **GAP-FILL** — the umbrella service that monitors your applications and automatically adjusts capacity to maintain steady, predictable performance at the lowest cost (covers EC2, ECS, DynamoDB, Aurora scaling).
- **AWS Compute Optimizer** **GAP-FILL** — uses ML to analyze historical utilization and recommend optimal AWS resource configurations (right-sizing recommendations) to reduce costs and improve performance.

- **AWS License Manager** **GAP-FILL** — track and manage software licenses (including BYOL) across AWS and on-prem.
- **AWS Service Catalog** **GAP-FILL** — lets organizations create and manage curated catalogs of approved IT services for self-service deployment by end users.
- **Service Quotas** **GAP-FILL** — view and manage your account's service limits (formerly "limits") from one console, and request increases.
- **AWS Control Tower** — best-practice multi-account setup with guardrails, built on top of Organizations.

SKIP — OUT OF SCOPE: AWS OpsWorks, AWS Chatbot, Amazon Data Lifecycle Manager, Elastic Transcoder, and Launch Wizard are out of scope.

Developer Tools

- **AWS CLI** — command-line interface for programmatic access to AWS services (contrast with the Management Console and SDKs — know the three access methods: console, CLI/SDK/API, Infrastructure as Code).
- **AWS CodeBuild** — fully managed CI build service; compiles code, runs tests, produces artifacts.
- **AWS CodePipeline** — fully managed CI/CD pipeline orchestrator.
- **AWS X-Ray** — distributed tracing; analyze and debug microservices and serverless apps.

SKIP — OUT OF SCOPE: AWS CodeDeploy, AWS CodeCommit (being retired), Amazon CodeGuru, AWS CodeStar, AWS CodeArtifact, AWS CloudShell, and AWS Device Farm are all out of scope for CLF-C02.

3.6 Application Integration

- **Amazon SQS** — queue (pull model), decouples services; Standard (best-effort ordering) vs FIFO (exactly-once, ordered, 3000 msg/s with batching). Messages retained up to 14 days.
- **Amazon SNS** — pub/sub (push model); push to many subscribers (email, SQS, Lambda, HTTP, SMS). Fan-out pattern: SNS → multiple SQS queues.
- **Amazon EventBridge** — serverless event bus; routes events from AWS services, SaaS apps, and custom apps to targets. Successor to CloudWatch Events.
- **AWS Step Functions** — orchestrates multi-step workflows as state machines; coordinates Lambda, ECS, DynamoDB, etc.

3.7 Analytics Services

Service	Purpose
Amazon Athena	Serverless SQL queries directly on S3 data; pay per query
Amazon Redshift	Cloud data warehouse; columnar; BI and analytics
Amazon EMR	Managed Hadoop/Spark/Hive cluster for big data processing
AWS Glue	Serverless ETL; data catalog; crawler discovers schema
Amazon QuickSight	Business Intelligence (BI) and data visualization; ML insights
Amazon OpenSearch Service	Managed search/log analytics (Elasticsearch-based)
Amazon Kinesis	Real-time streaming data (Data Streams / Firehose / Analytics)

3.8 Machine Learning & AI Services

Service	Capability
Amazon SageMaker AI	Build, train, and deploy ML models at scale

Amazon Rekognition	Image and video analysis (object/facial detection)
Amazon Transcribe	Speech-to-text
Amazon Translate	Neural machine translation
Amazon Polly	Text-to-speech
Amazon Comprehend	NLP: sentiment analysis, entity detection, key phrases
Amazon Lex	Build conversational chatbots (powers Alexa)
Amazon Kendra	ML-powered intelligent enterprise search engine
Amazon Textract	OCR + extract structured data from documents/forms
Amazon Q	Generative AI assistant for business; integrates with your data

SKIP — OUT OF SCOPE: Amazon Fraud Detector, Amazon Lookout for Metrics, Amazon Mechanical Turk, AWS Panorama, and Amazon Personalize are out of scope for CLF-C02.

3.9 Migration & Transfer

The 7 Rs of Migration

1. **Retire** — decommission unused apps.
2. **Retain** — keep on-premises (not ready to migrate).
3. **Rehost** (Lift & Shift) — move as-is to EC2; minimal change, fastest.
4. **Relocate** — move VMware workloads to VMware Cloud on AWS.
5. **Repurchase** — move to a different product (e.g. CRM → Salesforce).
6. **Replatform** (Lift, Tinker & Shift) — minor optimizations during migration (e.g. migrate DB to RDS).
7. **Refactor / Re-architect** — redesign cloud-native; highest effort, most benefit.
 - **AWS Application Migration Service (MGN)** — lift-and-shift; continuously replicates servers to AWS, then cutover.
 - **AWS Migration Hub** — central dashboard to track migration progress.
 - **AWS Application Discovery Service** — discovers on-prem servers, dependencies, and performance for migration planning.
 - **Migration Evaluator** **GAP-FILL** — builds a data-driven business case for AWS migration by analyzing your current on-prem environment (formerly "TSO Logic").
 - **AWS DMS / AWS SCT** — see Database Services above.

3.10 End-User Computing, Business Apps & Frontend

- **Amazon WorkSpaces** — managed cloud desktops (Virtual Desktop Infrastructure).
- **Amazon AppStream 2.0** — streams desktop applications to a browser; no installation needed.
- **Amazon WorkSpaces Secure Browser** **GAP-FILL** — fully managed, secure browser access to internal websites/SaaS apps without a full virtual desktop.
- **Amazon Connect** — cloud contact center (call center) solution.
- **Amazon SES (Simple Email Service)** **GAP-FILL** — scalable service for sending and receiving email, e.g. transactional or marketing email from applications.
- **AWS Amplify** **GAP-FILL** — set of tools/services to build and deploy full-stack web and mobile applications quickly, with built-in AWS backend integration.
- **AWS AppSync** — managed GraphQL service.
- **AWS IoT Core** — connect and manage billions of IoT devices; MQTT protocol.

SKIP — OUT OF SCOPE: Amazon WorkDocs, Amazon WorkMail, AWS Ground Station, and Amazon Chime are out of scope for CLF-C02 (despite sometimes appearing in older, non-official prep material).

4. Domain 4 — Billing, Pricing & Support

12%

Smallest domain, but every point here is an "easy" point if you drill it — the concepts are simple and repeat with predictable wording.

4.1 AWS Pricing Fundamentals

Three fundamentals of AWS pricing:

1. **Compute** — pay for the time resources run (per-second/hour billing).
2. **Storage** — pay for the amount of data stored (per GB/month).
3. **Data transfer** — inbound to AWS: **free**. Outbound to internet: charged. Between AZs: charged. Within the same AZ: free.

Free Tier

- **Always free** — e.g. Lambda 1M requests/month, DynamoDB 25 GB, CloudWatch 10 metrics.
- **12 months free** — e.g. EC2 750 hrs/month t2.micro, S3 5 GB, RDS 750 hrs db.t2.micro.
- **Trials** — short-term free trials for specific services.

4.2 Cost Management Tools

- **AWS Pricing Calculator** — estimate monthly costs *before* deploying anything.
- **AWS Cost Explorer** — visualize/understand/manage costs over time; 12 months historical data; forecasting; Reserved Instance recommendations.
- **AWS Budgets** — set custom cost/usage/RI/Savings Plan budgets and get alerts when thresholds are exceeded.
- **AWS Cost and Usage Report (CUR)** — the most comprehensive billing data; CSV to S3; pair with Athena/QuickSight for analysis.
- **AWS Cost Allocation Tags** — tag resources (e.g. `Project=Alpha`) to track costs by tag in billing reports.

4.3 Consolidated Billing & AWS Organizations

- **AWS Organizations** — create and manage multiple AWS accounts under one management account. Features: Consolidated Billing, SCPs, account vending (via Control Tower).
- **Consolidated Billing** — all linked accounts' usage aggregated; benefits from volume discounts and shared Reserved Instance/Savings Plan pricing across accounts.
- **Management Account** (formerly "master account") — pays for all linked accounts; SCPs cannot be applied to itself.
- **Service Control Policies (SCPs)** — applied at OU or account level; restrict max permissions; do **not** grant permissions.

EXAM TIP: Even with consolidated billing, each account's usage still appears separately in the Cost and Usage Report — but volume pricing is computed on the combined usage.

4.4 AWS Support Plans

Feature	Basic	Developer	Business	Enterprise
Cost	Free	\$29/mo	\$100/mo	\$15K/mo
Technical Account Manager (TAM)	No	No	No	Yes
24/7 phone/chat support	No	No	Yes	Yes
Business hours email	No	Yes	Yes	Yes

Trusted Advisor checks	7 checks	7 checks	All	All
Response: critical	—	—	1 hr	15 min
Response: production	—	—	4 hr	1 hr
Response: general	—	24 hr	24 hr	24 hr

AWS Enterprise On-Ramp — \$5,500/mo, a pool of TAMs, 30-min critical response (a step between Business and full Enterprise).

AWS Trust and Safety team **GAP-FILL** — the team you report abuse of AWS resources to (e.g. spam, malware hosted on AWS).

4.5 AWS Marketplace & Partner Network

- **AWS Marketplace** — digital catalog of thousands of software listings from independent software vendors (ISVs); buy, deploy, and manage third-party software on AWS. Pricing: free, BYOL, hourly, monthly, annual.
- **AWS Partner Network (APN)** — Consulting Partners (help design/build/migrate/manage workloads — SIs, MSPs) and Technology Partners (software/hardware integrated with AWS — ISVs, SaaS). Tiers: Select, Advanced, Premier.
- **Technical assistance beyond Support plans** — AWS Professional Services (paid consulting engagements) and AWS Solutions Architects (guidance during the sales/pre-sales process).
- **Free technical resources** — AWS Knowledge Center, AWS Prescriptive Guidance, AWS re:Post (community Q&A), AWS whitepapers and blogs.

5. Master Comparison Tables — The Pairs the Exam Loves to Confuse

5.1 Security Group vs. Network ACL

Feature	Security Group	Network ACL
Level	Instance/ENI	Subnet
Stateful/Stateless	Stateful	Stateless
Rules	Allow only	Allow and Deny
Default (new)	Deny all inbound, allow all outbound	Allow all inbound and outbound
Rule evaluation	All rules evaluated	Evaluated in order by rule number

5.2 CloudWatch vs. CloudTrail vs. Config

	CloudWatch	CloudTrail	AWS Config
What it does	Monitor metrics & logs	Audit API calls	Track config changes
Key question	Is the system healthy?	Who did what, when?	How did this change?
Main use	Ops monitoring, alerting	Compliance, forensics	Compliance, drift detection

5.3 RDS vs. Aurora vs. DynamoDB vs. Redshift

	RDS	Aurora	DynamoDB	Redshift
Type	Relational	Relational	NoSQL	Data Warehouse
Engine	6 options	MySQL/PG-compatible	Key-value	Columnar
Scaling	Vertical	Auto	Serverless	Node-based
Best for	OLTP	Managed OLTP	Millisecond latency	Analytics/BI

5.4 S3 vs. EBS vs. EFS

	S3	EBS	EFS
Type	Object	Block	File (NFS)
Access	Via HTTP/SDK	Single EC2 (mostly)	Multi-EC2 / multi-AZ
Size limit	Unlimited	Up to 64 TB	Unlimited
Region/AZ scope	Regional	Single AZ	Multi-AZ
OS	Any	Any	Linux only
Use case	Backup, web, static	Boot volumes, DBs	Shared storage

5.5 SQS vs. SNS vs. EventBridge

	SQS	SNS	EventBridge
Pattern	Queue (pull)	Pub/Sub (push)	Event bus (push)
Consumers	One consumer	Multiple	Multiple targets

Persistence	Up to 14 days	No retention	No retention
Use case	Decouple services	Alerts, fan-out	Event-driven architecture

5.6 Lambda vs. EC2 vs. ECS/Fargate

	Lambda	EC2	ECS/Fargate
Server management	None	You manage OS	None (Fargate)
Scaling	Automatic	Manual/ASG	Automatic
Runtime limit	15 minutes	Unlimited	Unlimited
Pricing	Per invocation	Per hour/second	Per task
Best for	Short, event-driven	Full control needed	Containers

5.7 CloudFront vs. Global Accelerator GAP-FILL

	CloudFront	Global Accelerator
What it optimizes	Content delivery (caches static/dynamic content)	Network path/routing performance
Protocol	HTTP/HTTPS	TCP and UDP
Best for	Websites, APIs, video/streaming	Gaming, VoIP, non-HTTP, multi-Region failover

6. Numbers & Limits to Memorize

- S3 max object size: **5 TB**
- S3 multipart upload recommended: **>100 MB** (required >5 GB)
- S3 Glacier Deep Archive retrieval: **12–48 hours**
- Lambda max timeout: **15 minutes**
- Lambda max memory: **10,240 MB (10 GB)**
- Lambda max deployment package: **50 MB (zipped)**
- EBS max size (single volume): **64 TB**
- SQS message retention: up to **14 days**
- SQS max message size: **256 KB**
- SQS FIFO throughput: **3,000 msg/s** with batching
- DynamoDB item max size: **400 KB**
- CloudTrail event retention (default): **90 days**
- CloudWatch Logs retention: configurable, **1 day – 10 years**
- Reserved Instance term: **1 or 3 years**
- Spot Instance savings: up to **90%**
- Reserved Instance / Savings Plan savings: up to **72%**
- Aurora storage copies: **6 copies across 3 AZs**
- RDS automated backup retention: up to **35 days**
- VPC default CIDR: **/16 max, /28 min**
- S3 durability: **99.999999999%** (11 nines)
- S3 Standard availability: **99.99%**
- IAM users per account (default): **5,000**
- Exam: **65 questions, 90 minutes, pass 700/1000**, 50 scored + 15 unscored

7. "Which Service Do I Use For...?" Cheat Sheet

Requirement	Service
Serverless functions	Lambda
Run containers without managing servers	Fargate
PaaS, deploy code without managing infra	Elastic Beanstalk
Host a static website cheaply	S3 Static Website Hosting
Global CDN, reduce latency	CloudFront
DNS management and routing	Route 53
Relational managed database	RDS or Aurora
NoSQL, single-digit ms latency	DynamoDB
Analytics / data warehouse	Redshift
Cache to reduce DB load	ElastiCache
Real-time data streaming	Kinesis
Decouple services (queue)	SQS
Push notifications / fan-out	SNS
Event-driven architecture	EventBridge
Dedicated private connection to AWS	Direct Connect
Encrypted tunnel over internet to AWS	Site-to-Site VPN
Connect many VPCs and on-prem	Transit Gateway
Private connectivity to AWS services (no internet)	PrivateLink / VPC Endpoints
Manage secrets / rotate passwords	Secrets Manager
Detect threats / unusual activity	GuardDuty
Find vulnerabilities in EC2	Inspector
Find PII in S3	Macie
Audit API calls	CloudTrail
Monitor metrics and set alarms	CloudWatch
Infrastructure as Code	CloudFormation
Migrate databases	DMS
Move large data offline	Snow Family (Snowball/Snowcone)
Download compliance reports	AWS Artifact
Visualize costs	Cost Explorer
Set spending alerts	AWS Budgets
Estimate AWS costs before using	Pricing Calculator
Multi-account management	AWS Organizations
ML image recognition	Rekognition

Text-to-speech	Polly
Speech-to-text	Transcribe
Managed Kubernetes	EKS
Managed Docker containers	ECS
Send transactional/marketing email	Amazon SES
Build a full-stack web/mobile app fast	AWS Amplify
Curated self-service catalog of approved services	AWS Service Catalog
Check/raise account service limits	Service Quotas
Right-size recommendations via ML	AWS Compute Optimizer

8. 20 Exam Traps & Wording Tricks

1. IAM is **global** — not tied to any Region.
2. S3 bucket names are **globally unique** — but buckets themselves are Region-specific.
3. Security Groups are **stateful** — allow inbound and the outbound reply is auto-allowed.
4. NACLs are **stateless** — you must define both inbound AND outbound rules.
5. VPC Peering is **not transitive** — need Transit Gateway for hub-and-spoke.
6. Spot Instances can be interrupted — **never** for databases or critical workloads.
7. RDS Multi-AZ is for **High Availability, not read performance** — use Read Replicas for reads.
8. CloudTrail logs **API calls**; CloudWatch logs **metrics and application logs**.
9. Shield Standard is **free and automatic** — Shield Advanced costs extra.
10. AWS Artifact is free and is a **compliance document portal**, not a security scanning service.
11. Trusted Advisor has full checks only for **Business and Enterprise** support plans.
12. A Technical Account Manager (TAM) is **Enterprise support only**.
13. NAT Gateway sits in a **public** subnet but serves **private** subnet instances.
14. S3 durability is 11 nines — **availability varies by storage class**, durability does not.
15. Reserved Instances: Standard (sellable on RI Marketplace) vs. Convertible (exchange for a different instance type).
16. Savings Plans are more flexible than RIs — you commit to **\$/hour**, not a specific instance type.
17. Gateway Endpoints (S3, DynamoDB) are **free**; Interface Endpoints cost money.
18. ACM certificates are **free** for use with AWS services (CloudFront, ALB).
19. Data transfer **IN** to AWS is always free; data transfer **OUT** costs money.
20. Watch the qualifier words: "MOST cost-effective," "LEAST operational overhead," "MOST secure" — each one changes which answer is correct, even with identical services listed.

9. Explicitly Out-of-Scope — Stop Reading About These

These names might appear in older study materials or in distractor answers on the real exam, but the official CLF-C02 exam guide lists them as out of scope. If you see them as an answer option, that is often a strong signal it is the *wrong* option.

- AWS App Runner, AWS Copilot, AWS Wavelength
- Amazon AppFlow, AWS Clean Rooms, AWS Data Exchange, Amazon DataZone, Amazon MSK, Amazon Timestream
- AWS AppFabric, Amazon Simple Workflow Service
- Amazon WorkDocs, Amazon WorkMail
- AWS Application Cost Profiler, Amazon DevPay, AWS Billing Conductor
- AWS Activate, AWS IQ, AWS Managed Services (AMS)
- Amazon Keyspaces, Amazon MemoryDB for Redis, AWS AppConfig
- AWS Application Composer, AWS CodeArtifact, AWS CodeDeploy, Amazon CodeGuru, AWS CloudShell, AWS Device Farm
- Amazon GameLift, Amazon Lumberyard (Game Tech)
- AWS IoT Device Defender, AWS IoT Greengrass, Amazon Monitron
- Amazon Fraud Detector, Amazon Lookout for Metrics, Amazon Mechanical Turk, AWS Panorama, Amazon Personalize
- AWS Chatbot, Amazon Data Lifecycle Manager, Amazon Elastic Transcoder, AWS Launch Wizard
- All AWS Elemental media services, Amazon Interactive Video Service
- AWS Migration Hub Refactor Spaces, AWS Transfer Family
- AWS Cloud Map, AWS Network Access Analyzer, AWS Ground Station, Amazon VPC Lattice
- Amazon Cloud Directory, AWS Network Firewall
- AWS RoboMaker
- Amazon FSx for Lustre

10. Exam Day Logistics & Strategy

1. **Eliminate wrong answers first.** Usually 2 options are obviously wrong; choose between the remaining 2.
2. **Watch for qualifiers:** MOST cost-effective, LEAST operational overhead, MOST secure — each changes the answer.
3. **No servers to manage** ⇒ **Serverless** (Lambda, Fargate, Aurora Serverless, DynamoDB).
4. **High availability** ⇒ Multi-AZ, Auto Scaling, multiple Regions.
5. **Decouple** ⇒ SQS (queue) or SNS (pub/sub).
6. **Compliance reports** ⇒ AWS Artifact (not GuardDuty, not Config).
7. **Detect threats** ⇒ GuardDuty; **block web attacks** ⇒ WAF; **DDoS** ⇒ Shield.
8. **Audit API calls** ⇒ CloudTrail; **monitor metrics** ⇒ CloudWatch; **config changes** ⇒ Config.
9. **Migrate a database** ⇒ DMS; **convert schema** ⇒ SCT.
10. Flag and skip difficult questions; return at the end.
11. 90 minutes for 65 questions ≈ **83 seconds/question**. Keep moving — don't let one question eat your buffer.
12. If unsure, go with your gut, then mark for review rather than spending too long. No penalty for guessing, so **never leave a question blank**.
13. Both multiple-choice (1 of 4) and multiple-response (2+ of 5+) questions appear — read the instructions on each question carefully to know how many to select.

EXAM TIP: 15 of the 65 questions are unscored and indistinguishable from scored ones. You cannot tell which — treat every question with equal seriousness.

11. Final Gap-Fill Additions

These surfaced from a later cross-check against additional source material. Read this section once — ten minutes — before the self-test.

11.1 New In-Scope Items to Lock In

IAM Access Advisor

Shows which AWS services a user/role has **actually used**, and when. Complements the IAM Credential Report (which shows password/MFA/key status). Use case: tightening least privilege by spotting permissions nobody actually uses.

EC2 Image Builder

Automates building, testing, and distributing your own AMIs. One-liner: "AMI creation, automated and repeatable" instead of manually creating golden images each time.

S3 Transfer Acceleration

Speeds up uploads/downloads to S3 by routing traffic through CloudFront's edge network instead of the public internet directly. Different from Multipart Upload (which splits large files) — Transfer Acceleration is about the network path, not file splitting.

Root User's Exclusive Privileges — memorize this exact list

EXAM TIP: Only the root user can: change account settings · close the AWS account · change or cancel the Support plan · register as a seller in the Reserved Instance Marketplace.

Billing Alarms Live in us-east-1

CloudWatch billing metrics are only available in the **us-east-1 (N. Virginia)** Region — regardless of which Region your actual resources run in. If a question describes a billing alarm that "isn't showing up," the fix is often "check you're looking in us-east-1."

Cost Anomaly Detection vs. AWS Budgets

	AWS Budgets	Cost Anomaly Detection
How it works	You set a threshold; alerts when crossed	ML learns your normal spend pattern; alerts on unusual deviations
Catches	"You went over \$2,000"	"Your spend looks weird even though you're still under budget"

AWS STS (Security Token Service)

Issues temporary, limited-privilege credentials. This is the mechanism *underneath* IAM Roles — when you assume a role, STS is what actually hands you the temporary credentials. One-liner: **STS = the engine that generates temporary credentials; a Role = the thing you assume to get them.**

Disaster Recovery Strategy Tiers — memorize this table

Ordered from cheapest/slowest recovery to most expensive/fastest recovery. This exact ordering is a common exam question shape.

Strategy	Recovery Time (RTO)	Relative Cost	What it means
Backup & Restore	Hours	\$	Just back up data; restore from scratch when disaster hits
Pilot Light	Minutes–hours	\$\$	Core systems always running at minimal scale, scale up on failover

Warm Standby	Minutes	\$\$\$	Scaled-down but fully functional copy always running
Multi-site / Hot-site	Near-zero	\$\$\$\$	Full production-scale copy running in another Region, active-active

High Availability vs. Scalability vs. Elasticity vs. Agility GAP-FILL

Term	Means
High Availability	System stays up despite failures (Multi-AZ)
Scalability	Can grow to handle more load (vertical or horizontal)
Elasticity	Scales automatically up and down with demand
Agility	How fast you can provision/change resources

Compromised Account Sequence — memorize verbatim

EXAM TIP: If your AWS account is compromised: change the root password → delete and rotate all IAM users' passwords/access keys → contact AWS Support. This exact sequence is a realistic exam scenario answer.

11.2 Additional Names to Recognize Only — Do NOT Study Details

These appear in some third-party study material but are **not** on the official CLF-C02 in-scope list. If you see one as an answer option, that's often a signal it's the wrong option (a distractor pulled from a higher-level exam).

- Amazon QLDB, Amazon Managed Blockchain
- Amazon MQ, AWS Network Firewall, Amazon CodeGuru
- AWS CodeDeploy, CodeCommit, CodeArtifact, CodeStar, Cloud9, AWS CDK, AWS OpsWorks
- Amazon Forecast, Amazon Personalize
- Amazon Elastic Transcoder, AWS Device Farm
- AWS DataSync, AWS Fault Injection Simulator (FIS)
- AWS Ground Station, Amazon Pinpoint

EXAM TIP: If the exam shows any of the "recognize only" names above as an answer choice next to a name from this guide's in-scope sections, lean toward the in-scope name — CLF-C02 questions are built from the official content outline, and out-of-scope services mostly show up as distractors.

12. Extra Concepts From Practice Exams & Tonight's Tests

Pulled from the four uploaded practice exams (Exams 2, 8, 10, 11), the Medium cheat sheet, and the specific questions you missed across tonight's tests. Read this section last, right before sleep — it's the final 10% that rounds everything out.

12.1 Reserved Instance Payment Options — the full picture

Beyond Standard vs. Convertible, RIs also vary by **how much you pay upfront**. This combination (term + payment option) is what determines the exact discount level, and practice exams test this combination directly.

Payment Option	How it works	Relative Discount
No Upfront	Pay nothing upfront; billed hourly over the term	Lowest discount
Partial Upfront	Pay part upfront; remainder billed hourly	Medium discount
All Upfront	Pay the full term cost upfront	Highest discount

EXAM TIP: The deepest possible RI discount = **3-year term + All Upfront + Standard RI**. Every extra bit of commitment (longer term, more upfront, less flexibility) buys a bigger discount. Convertible RIs trade some discount for the ability to change instance families later.

12.2 AWS Support Plans — the extra pieces

Your Master Guide's Support Plans table (Section 4.4) is correct — these three items appeared repeatedly across the practice exams and are worth calling out explicitly:

- **AWS Concierge (Support Concierge Service)** — Enterprise-only team that helps specifically with billing and account inquiries, distinct from technical support.
- **Infrastructure Event Management (IEM)** — Enterprise-only proactive guidance for planned events (e.g. a big product launch or expected traffic surge) — architectural and scaling advice ahead of time, not reactive troubleshooting.
- **AWS Support API** — lets you programmatically create and manage support cases; available from the Business plan upward.

12.3 Health Dashboards — don't mix these up

	AWS Service Health Dashboard	AWS Personal Health Dashboard
Scope	Public, general status of all AWS services across all Regions	Private, personalized to YOUR account's specific resources
Use case	"Is S3 down anywhere right now?"	"Is anything currently affecting MY resources specifically?"

12.4 Credentials Cheat Sheet — which credential does which job

Credential Type	Used for
Username + Password	Logging into the AWS Management Console (human, browser-based)
Access Key ID + Secret Access Key	Programmatic access via CLI, SDK, or direct API calls
Key Pair (SSH key)	Logging into an EC2 instance's operating system directly via SSH
MFA Token	Second-factor verification alongside a password or during sensitive actions
Temporary credentials (via STS)	Issued when a role is assumed — never used for a permanent IAM user

EXAM TIP: Only IAM Users and the root user can have long-lived access keys. IAM Roles never have static access keys — they only ever get temporary credentials via STS.

12.5 AWS Quick Start Reference Deployments

Pre-built, automated reference architectures for deploying popular third-party technologies (e.g. IBM MQ, SAP, Microsoft SQL Server) onto AWS with minimal effort and time. If a question asks "how do I deploy [well-known enterprise software] on AWS quickly," Quick Starts is often the intended answer.

12.6 Legacy or Out-of-Scope Names You'll See as Distractors

These appeared as wrong-answer options in the practice exams. They're either legacy/superseded services or outside official CLF-C02 scope — recognize them only to correctly rule them out:

- **Amazon SWF (Simple Workflow Service)** — legacy workflow orchestration, largely superseded by AWS Step Functions.
- **AWS Server Migration Service** — legacy, superseded by AWS Application Migration Service (MGN).
- **Amazon Machine Learning** — a retired, legacy ML service, superseded by Amazon SageMaker.
- **Amazon Elastic Transcoder / AWS Elemental MediaConvert** — media transcoding services; not on the official CLF-C02 in-scope list despite appearing in some prep material.
- **Amazon Kinesis Video Streams** — a real, narrower Kinesis offering for video specifically; not typically the intended answer for general streaming/CDN questions (that's usually CloudFront or Kinesis Data Streams).

12.7 Five Final Corrections — from tonight's last test

These are the exact concepts behind your final five missed questions. Read each once before sleeping.

Service	What it actually does	Don't confuse with
VPC Endpoint	Private connection from your VPC to a supported AWS service (e.g. S3), without using the internet	PrivateLink (connects to any service, including third-party/your own, sitting in another VPC) and VPC Peering (connects two VPCs to each other, not to a service)
Amazon EMR	Managed Hadoop/Spark/Hive cluster — you're running a big data processing engine	AWS Glue (serverless ETL, no cluster to manage) and Amazon Athena (SQL queries directly on S3, no cluster or ETL)
AWS TCO Calculator	Compares on-premises cost vs. AWS cost — used for migration business cases	AWS Pricing Calculator (estimates AWS-only cost for a planned architecture, no on-prem comparison)
Amazon Lightsail	Simple apps, fixed predictable monthly price, bundles compute+storage+networking+DNS together	EC2 (more flexible/configurable, but requires assembling the pieces yourself)
Elastic Transcoder / MediaConvert	Real services, but NOT on the official CLF-C02 in-scope list	Don't pick these even when they sound like the "obviously correct" technical answer — check scope first

12.8 Aggregate List — Services Most Often Confused Across All Your Tests Tonight

Across every test and drill tonight, these specific pairs/groups caused the most repeated misses. This is your final "read twice" list:

- On-Demand vs. Reserved vs. Savings Plans vs. Spot vs. Dedicated Host/Instance
- AWS Support Plans vs. TAM vs. Professional Services vs. Solutions Architects vs. Training options
- Pricing Calculator vs. TCO Calculator vs. Cost Explorer vs. CUR vs. Cost Anomaly Detection
- Security Group vs. Network ACL
- CloudWatch vs. CloudTrail vs. Config

- RDS Multi-AZ vs. Read Replicas
- VPC Endpoint vs. PrivateLink vs. VPC Peering vs. Transit Gateway
- EMR vs. Glue vs. Athena vs. Redshift
- Multi-select technique: eliminate absolute-sounding options first ("always," "never," "guarantees," "no responsibility for...")

EXAM TIP: If you only have time to reread one thing tomorrow morning, reread this list. Every one of these pairs showed up as a real miss at least once across tonight's five tests — this is genuinely your highest-leverage five minutes.

13. Closed-Book 50-Question Self-Test

Rules: Cover the answer key. Do not look back at earlier pages. Time yourself: 50 minutes max (this mirrors real exam pace). Weighted like the real exam: Q1–12 = Domain 1, Q13–27 = Domain 2, Q28–44 = Domain 3, Q45–50 = Domain 4. Answer key with one-line rationale is at the end of this section.

Domain 1 — Cloud Concepts (Q1–12)

Q1. A company wants to eliminate large upfront hardware purchases and pay only for what it uses. Which cloud advantage does this describe?

- A) Go global in minutes B) Trade CapEx for OpEx C) Stop guessing capacity D) Economies of scale

Q2. Which AWS Global Infrastructure component is used by CloudFront to cache content close to end users?

- A) Availability Zone B) Region C) Edge Location D) Local Zone

Q3. A company wants to run legacy software using an existing on-prem license on AWS. Which purchasing/licensing option supports this?

- A) Reserved Instances B) BYOL on a Dedicated Host C) Spot Instances D) Savings Plans

Q4. Which service model requires the customer to manage the OS, runtime, and applications, while AWS manages the physical infrastructure?

- A) SaaS B) PaaS C) IaaS D) FaaS

Q5. (Select TWO) Which are true about Availability Zones?

- A) They share single points of failure B) They are connected by low-latency links C) A Region has at least 2 AZs D) An AZ can span multiple Regions E) AZs are only used by CloudFront

Q6. Which pillar of the Well-Architected Framework focuses on eliminating unnecessary spending through right-sizing and pricing models?

- A) Operational Excellence B) Reliability C) Cost Optimization D) Performance Efficiency

Q7. A company needs its application to survive the failure of an entire data center. What should it do?

- A) Deploy to a single AZ with a larger instance B) Deploy across multiple AZs C) Increase EBS volume size D) Enable S3 versioning

Q8. Which AWS CAF benefit is described as "the ability to reduce risk from security incidents and compliance breaches"?

- A) Increased revenue B) Reduced business risk C) Increased operational efficiency D) Improved ESG performance

Q9. A retail company wants to expand into three new countries with minimal delay. Which cloud benefit supports this best?

- A) Stop guessing capacity B) Go global in minutes C) Trade CapEx for OpEx D) Economies of scale

Q10. Which deployment model connects an on-premises data center to the AWS Cloud via VPN or Direct Connect?

A) Public cloud B) Private cloud C) Hybrid cloud D) Multi-cloud

Q11. Under the Well-Architected Framework, which pillar is concerned with minimizing environmental impact?

A) Reliability B) Sustainability C) Security D) Performance Efficiency

Q12. Which migration strategy involves moving an application to AWS with almost no code changes?

A) Refactor B) Repurchase C) Rehost D) Retire

Domain 2 — Security & Compliance (Q13–27)

Q13. Under the Shared Responsibility Model, who is responsible for patching the guest OS on an EC2 instance?

A) AWS B) The customer C) Both equally D) Neither — it's automatic

Q14. Which service should a company use to store and automatically rotate a database password used by an application?

A) AWS KMS B) AWS Secrets Manager C) AWS Certificate Manager D) Amazon Cognito

Q15. A company wants free, automatic DDoS protection with no extra configuration. Which service provides this by default?

A) AWS WAF B) AWS Shield Advanced C) AWS Shield Standard D) AWS Firewall Manager

Q16. Which IAM policy evaluation rule is correct?

A) An explicit Allow always overrides a Deny B) An explicit Deny always overrides an Allow C) Implicit allow is the default D) SCPs grant permissions directly

Q17. Where can a customer download AWS's SOC 2 compliance report?

A) AWS Trusted Advisor B) AWS Artifact C) AWS Config D) Amazon Inspector

Q18. Which service uses machine learning to analyze CloudTrail, VPC Flow Logs, and DNS logs to detect malicious activity?

A) Amazon Macie B) Amazon GuardDuty C) AWS Config D) Amazon Inspector

Q19. Which service discovers personally identifiable information (PII) stored in Amazon S3?

A) Amazon Macie B) AWS Inspector C) AWS Config D) Amazon GuardDuty

Q20. A company wants a single sign-on experience across multiple AWS accounts and business applications. What should they use?

A) IAM Users B) IAM Identity Center C) AWS Organizations D) Amazon Cognito

Q21. Which logging service records every API call made in an AWS account?

A) Amazon CloudWatch B) AWS CloudTrail C) AWS Config D) VPC Flow Logs

Q22. (Select TWO) Which are best practices for protecting the AWS root user?

A) Use it for daily administrative tasks B) Enable MFA C) Share credentials with the whole ops team D) Avoid daily use, use IAM users/roles instead E) Disable IAM entirely

Q23. Which AWS service lets you build a custom Hardware Security Module (HSM) that you fully manage the keys for, inside your VPC?

A) AWS KMS B) AWS CloudHSM C) AWS Secrets Manager D) ACM

Q24. A company needs to track configuration changes to its AWS resources over time and evaluate them against compliance rules. Which service?

A) AWS CloudTrail B) AWS Config C) Amazon CloudWatch D) AWS Trusted Advisor

Q25. Which of the following authenticates users for a customer-facing mobile app, providing user pools and identity pools?

A) IAM Identity Center B) Amazon Cognito C) AWS Directory Service D) AWS IAM

Q26. Which service centrally manages AWS WAF rules, AWS Shield Advanced protections, and security groups across multiple accounts?

A) AWS Security Hub B) AWS Firewall Manager C) Amazon Detective D) AWS Organizations

Q27. A regulated company must continuously assess its AWS usage against industry compliance frameworks. Which service is purpose-built for this?

A) AWS Trusted Advisor B) AWS Audit Manager C) Amazon Detective D) AWS Config

Domain 3 — Cloud Technology & Services (Q28–44)

Q28. Which EC2 purchasing option offers the deepest discount but can be interrupted by AWS with short notice?

A) Reserved Instance B) Savings Plan C) Spot Instance D) Dedicated Host

Q29. Which storage class offers the lowest cost for data that is almost never accessed and can tolerate a 12–48 hour retrieval time?

A) S3 Standard-IA B) S3 One Zone-IA C) S3 Glacier Deep Archive D) S3 Intelligent-Tiering

Q30. A company needs a shared file system accessible by many Linux EC2 instances across multiple AZs simultaneously. Which service?

A) Amazon EBS B) Amazon EFS C) Instance Store D) Amazon FSx for Windows

Q31. Which database service is best suited to an application requiring single-digit-millisecond latency at unlimited scale, with a flexible (non-relational) schema?

A) Amazon RDS B) Amazon Redshift C) Amazon DynamoDB D) Amazon Aurora

Q32. A company wants automatic failover to a standby database in a second AZ for high availability. What should they enable?

A) RDS Read Replica B) RDS Multi-AZ C) DynamoDB Global Tables D) ElastiCache

Q33. Which load balancer type operates at Layer 7 and can route based on URL path or hostname?
A) Network Load Balancer B) Gateway Load Balancer C) Application Load Balancer D) Classic Load Balancer

Q34. Which service lets private subnet EC2 instances initiate outbound internet connections without accepting inbound ones?
A) Internet Gateway B) NAT Gateway C) VPC Peering D) Transit Gateway

Q35. Which of the following is a stateless, subnet-level firewall requiring explicit inbound AND outbound rules?
A) Security Group B) Network ACL C) AWS WAF D) Firewall Manager

Q36. A company wants to connect thousands of VPCs and on-premises networks using transitive routing through a single hub. Which service?
A) VPC Peering B) AWS Transit Gateway C) AWS Direct Connect D) NAT Gateway

Q37. Which serverless service lets developers run code without provisioning or managing servers, billed per invocation?
A) Amazon EC2 B) AWS Lambda C) Amazon Lightsail D) AWS Elastic Beanstalk

Q38. Which service decouples microservices using a pull-based message queue with up to 14 days of message retention?
A) Amazon SNS B) Amazon SQS C) Amazon EventBridge D) AWS Step Functions

Q39. A company wants to run serverless SQL queries directly against data stored in S3 without standing up a database. Which service?
A) Amazon Redshift B) Amazon Athena C) Amazon RDS D) AWS Glue

Q40. Which AWS ML service converts spoken audio into text?
A) Amazon Polly B) Amazon Transcribe C) Amazon Translate D) Amazon Comprehend

Q41. A company needs to migrate a large on-prem Oracle database to Amazon Aurora, converting the schema in the process. Which two services are used together?
A) DMS and Snowball B) DMS and AWS SCT C) Migration Hub and CloudEndure D) Application Discovery Service and MGN

Q42. Which service provides a managed Kubernetes control plane?
A) Amazon ECS B) Amazon EKS C) AWS Fargate D) AWS Batch

Q43. Which service streams full desktop applications to a browser without installing anything locally?
A) Amazon WorkSpaces B) Amazon AppStream 2.0 C) Amazon Chime D) AWS Client VPN

Q44. Which AWS service would a company use to send transactional order-confirmation emails to customers at scale?
A) Amazon SNS B) Amazon Connect C) Amazon SES D) Amazon Pinpoint

Domain 4 — Billing, Pricing & Support (Q45–50)

Q45. Which tool should a company use to estimate the monthly cost of an architecture before deploying it?
A) AWS Cost Explorer B) AWS Pricing Calculator C) AWS Budgets D) AWS Cost and Usage Report

Q46. Which AWS Support plan includes a dedicated Technical Account Manager (TAM)?
A) Developer B) Business C) Enterprise D) Basic

Q47. Data transfer from the internet INTO an AWS account is billed how?
A) Always free B) Always charged C) Free only under 1GB D) Charged only for EC2

Q48. Which billing feature lets multiple linked AWS accounts share volume discounts and Reserved Instance benefits under one management account?
A) AWS Budgets B) Consolidated Billing C) Cost Allocation Tags D) AWS Marketplace

Q49. Which of the following restricts the maximum permissions available to accounts in an AWS Organization but does not itself grant any permissions?
A) IAM Policy B) Permission Boundary C) Service Control Policy (SCP) D) Resource-based Policy

Q50. A startup wants to purchase pre-built, third-party security software directly usable on AWS with flexible pricing (hourly, monthly, BYOL). Where would they go?
A) AWS Partner Network B) AWS Marketplace C) AWS Artifact D) AWS Service Catalog

Answer Key & Rationale

Q	Ans	Why
1	B	Pay only for what you use, no upfront investment = trading CapEx for OpEx.
2	C	Edge Locations are used by CloudFront/Route 53 to cache content near users.
3	B	Dedicated Hosts support BYOL because you control the physical server for licensing purposes.
4	C	IaaS: AWS manages physical infra, customer manages OS/runtime/apps.
5	B, C	AZs are low-latency connected and don't share failure points; a Region has 2+ AZs.
6	C	Cost Optimization pillar = eliminate waste, right-size, use pricing models.
7	B	Multi-AZ deployment protects against loss of an entire data center (single AZ).
8	B	AWS CAF benefit explicitly named "reduced business risk."
9	B	"Go global in minutes" = deploying to new Regions quickly.
10	C	Hybrid cloud = on-prem/private + public cloud connected via VPN/Direct Connect.
11	B	Sustainability pillar = minimize environmental impact.
12	C	Rehost ("lift and shift") = move as-is, minimal changes, fastest migration.
13	B	For EC2, the customer patches the guest OS; AWS only patches OS for managed services like RDS/Lambda.
14	B	Secrets Manager stores AND automatically rotates credentials; KMS just manages encryption keys.
15	C	Shield Standard is free and automatic on every account; Advanced is paid.
16	B	Explicit Deny always wins. Default is implicit deny, not allow. SCPs restrict, they don't grant.
17	B	AWS Artifact is the self-service compliance report/document portal.
18	B	GuardDuty is the ML-based threat detection service analyzing those exact log sources.
19	A	Macie uses ML specifically to find sensitive data/PII in S3.
20	B	IAM Identity Center (formerly AWS SSO) is built for cross-account, cross-app single sign-on.
21	B	CloudTrail logs every API call (who did what, when).
22	B, D	MFA + avoiding daily root use (delegate to IAM users/roles) are core root-protection practices.
23	B	CloudHSM = dedicated, customer-managed HSM in your VPC. KMS is AWS-managed keys, not a dedicated HSM you control directly.
24	B	AWS Config tracks configuration changes and evaluates compliance rules over time.
25	B	Cognito provides user pools (authentication) and identity pools (authorization) for app users.
26	B	Firewall Manager centrally manages WAF, Shield Advanced, and security groups across accounts.
27	B	Audit Manager is purpose-built to continuously assess AWS usage against compliance frameworks.
28	C	Spot Instances offer up to 90% savings but can be interrupted.
29	C	S3 Glacier Deep Archive: cheapest, 12–48 hr retrieval, for rarely accessed long-term archives.
30	B	EFS is a managed NFS file system shared across many Linux instances/AZs; EBS is single-instance/single-AZ.
31	C	DynamoDB: single-digit ms latency, unlimited scale, NoSQL/flexible schema.
32	B	RDS Multi-AZ = synchronous standby + automatic failover = high availability, not read scaling.

33	C	ALB = Layer 7, path/host-based routing.
34	B	NAT Gateway allows outbound-only internet access for private subnet resources.
35	B	NACLs are stateless and subnet-level, requiring explicit inbound and outbound rules.
36	B	Transit Gateway is the hub-and-spoke, transitive-routing solution for many VPCs/on-prem networks.
37	B	Lambda = no server provisioning, billed per invocation/duration.
38	B	SQS is the pull-based queue with up to 14-day retention.
39	B	Athena = serverless SQL directly on S3 data, pay per query.
40	B	Transcribe = speech-to-text (Polly is the opposite: text-to-speech).
41	B	DMS migrates the data; SCT converts the schema between different engines (Oracle → Aurora).
42	B	EKS = managed Kubernetes control plane.
43	B	AppStream 2.0 streams applications to a browser; WorkSpaces provides a full virtual desktop instead.
44	C	Amazon SES is purpose-built for sending transactional/marketing email at scale.
45	B	Pricing Calculator estimates cost BEFORE you deploy; Cost Explorer analyzes cost AFTER usage exists.
46	C	TAM is exclusive to the Enterprise support plan (and Enterprise On-Ramp).
47	A	Inbound data transfer to AWS is always free.
48	B	Consolidated Billing aggregates usage across linked accounts for shared volume/RI discounts.
49	C	SCPs set a maximum permission ceiling; they never grant permissions themselves.
50	B	AWS Marketplace is the catalog for buying/deploying third-party software with flexible pricing.

EXAM TIP: Scoring guide: 45–50 correct → you're ready, book the exam. 38–44 → solid, but re-read the domain(s) where you missed the most before booking. Below 38 → do one more focused pass tonight on your weakest domain, then retake just those questions.

Final word: Every fact in this document was cross-checked against your two source PDFs and the structure of the official CLF-C02 exam guide. You do not need any other resource today. Work the schedule, take the test honestly with the answers covered, fix your gaps, and go book the exam. Good luck — you've got this.